

Polar Curves and Derivatives

ANSWER KEY



Converting between polar and rectangular coordinates

- 1 Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.

$$x = r \cos \theta = 4 \cos \left(\frac{\pi}{3}\right) = 4\left(\frac{1}{2}\right) = 2. \quad y = r \sin \theta = 4 \sin \left(\frac{\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}. \quad \text{Point: } (2, 2\sqrt{3}).$$

- 2 Convert the polar equation $r = 4 \cos \theta$ to rectangular form, and identify the curve.

Multiply by r : $r^2 = 4r \cos \theta \Rightarrow x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0 \Rightarrow (x - 2)^2 + y^2 = 4$. This is a circle of radius 2 centered at $(2, 0)$.

Identifying and sketching common polar curves

- 3 Identify the type of curve for $r = 2 + 2 \sin \theta$, and state its key feature.

This is a cardioid (since the coefficient of $\sin \theta$ equals the constant term) — a heart-shaped curve passing through the pole.

- 4 Identify the type of curve for $r = \cos(3\theta)$.

This is a rose curve with 3 petals (odd n in $r = \cos(n\theta)$ gives n petals).

The slope of a polar curve

- 5 State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r \cos \theta$ and $y = r \sin \theta$.

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}.$$

- 6 Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.

$$\frac{dr}{d\theta} = 2. \quad x = 2\theta \cos \theta, \quad y = 2\theta \sin \theta. \quad \frac{dy}{d\theta} = 2 \sin \theta + 2\theta \cos \theta; \quad \frac{dx}{d\theta} = 2 \cos \theta - 2\theta \sin \theta. \quad \text{At } \theta = \frac{\pi}{2}: \\ \frac{dy}{d\theta} = 2(1) + 2\left(\frac{\pi}{2}\right)(0) = 2; \quad \frac{dx}{d\theta} = 2(0) - 2\left(\frac{\pi}{2}\right)(1) = -\pi. \quad \frac{dy}{dx} = \frac{2}{-\pi} = -\frac{2}{\pi}.$$

Finding horizontal and vertical tangents

- 7 For the circle $r = 4 \cos \theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r \sin \theta$ has a critical point).

$$y = 4 \cos \theta \sin \theta = 2 \sin(2\theta). \quad \frac{dy}{d\theta} = 4 \cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}.$$

Symmetry of polar curves

- 8 Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x -axis).

Replace θ with $-\theta$ in the equation. If the equation is unchanged (or yields the same curve), the curve is symmetric about the polar axis. For example, $r = 2 + 2\cos \theta$ satisfies $f(-\theta) = 2 + 2\cos(-\theta) = 2 + 2\cos \theta = f(\theta)$, confirming symmetry.

- 9 Test whether $r = 3\sin \theta$ is symmetric about the y -axis (replace θ with $\pi - \theta$).

$f(\pi - \theta) = 3\sin(\pi - \theta) = 3\sin \theta = f(\theta)$ (using $\sin(\pi - \theta) = \sin \theta$). Symmetric about the y -axis.

More polar-to-rectangular conversions

- 10 Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \leq \theta < 2\pi$).

$r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$. $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ (in Quadrant I, matching both coordinates positive).

- 11 Convert the rectangular equation $y = x$ to polar form.

$r\sin \theta = r\cos \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ (a constant- θ ray, i.e. a straight line through the pole).

- 12 Convert the polar equation $r\sin \theta = 4$ to rectangular form, and identify the curve.

$r\sin \theta = y$, so $y = 4$: a horizontal line.

More slope-of-a-polar-curve practice

- 13 Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin \theta$ at $\theta = 0$.

$\frac{dr}{d\theta} = \cos \theta$. $\frac{dy}{dx} = \frac{\cos \theta \sin \theta + r \cos \theta}{\cos \theta \cos \theta - r \sin \theta}$. At $\theta = 0$: $r = 1$, $\frac{dr}{d\theta} = 1$. Numerator: $1(0) + 1(1) = 1$. Denominator: $1(1) - 1(0) = 1$. $\frac{dy}{dx} = \frac{1}{1} = 1$.

- 14 Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ , and interpret the result using the fact that a circle centered at the origin has radial symmetry.

$\frac{dr}{d\theta} = 0$. $\frac{dy}{dx} = \frac{0 \cdot \sin \theta + 5 \cos \theta}{0 \cdot \cos \theta - 5 \sin \theta} = \frac{5 \cos \theta}{-5 \sin \theta} = -\cot \theta$. This matches the slope of the tangent line to a circle at angle θ — perpendicular to the radius at that point, consistent with basic circle geometry.

Tangent lines at the pole

- 15 Find the values of θ where the cardioid $r = 1 - \cos \theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

$1 - \cos \theta = 0 \Rightarrow \cos \theta = 1 \Rightarrow \theta = 0$. At the single angle where $r = 0$, the curve touches the pole; the tangent line there is along $\theta = 0$ (the polar axis), since that's the direction the curve approaches the origin from.

- 16 Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.

$\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — these are the four tangent-line directions at the pole for this 4-petaled rose.

Classifying polar curves by equation type

- 17 Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

a $r = 5$ Circle centered at the pole

b $r = 3 + 3\cos \theta$ Cardioid (coefficient equals constant)

c $r = 1 + 3\sin \theta$ Limaçon with an inner loop (constant < coefficient)

d $r = 4\sin(5\theta)$ Rose with 5 petals (odd $n = 5$)

Number of petals in rose curves

- 18 State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.

If n is odd, the rose has exactly n petals (traced over $[0, \pi]$). If n is even, the rose has $2n$ petals (requires the full $[0, 2\pi]$ to trace, since each petal is retraced with opposite sign after π).

- 19 How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?

$r = 2\sin(4\theta)$: $n = 4$ (even) $\Rightarrow 2(4) = 8$ petals. $r = 5\cos(7\theta)$: $n = 7$ (odd) $\Rightarrow 7$ petals.

Graphing polar curves point-by-point

- 20 Complete a short table of (θ, r) values for $r = 2 + 2\cos \theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

$\theta = 0$: $r = 4$. $\theta = \frac{\pi}{2}$: $r = 2$. $\theta = \pi$: $r = 0$. $\theta = \frac{3\pi}{2}$: $r = 2$. The curve is widest ($r = 4$) along the polar axis, pinches to a point ($r = 0$) at $\theta = \pi$ (opposite direction), forming the characteristic heart shape.

Concavity considerations for polar curves

- 21 Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).

A vertical tangent means the slope $\frac{dy}{dx}$ is undefined (infinite) — since $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$, this happens exactly when the denominator $\frac{dx}{d\theta} = 0$ while the numerator stays nonzero, making the ratio blow up.

Multiple choice — easy 15

- 22 How many petals does the rose $r = \cos(2\theta)$ have?

- a. 4 ✓
- b. 6
- c. 7
- d. 2

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

- 23 How many petals does the rose $r = 7\cos(6\theta)$ have?

- a. 14
- b. 10
- c. 6
- d. 12 ✓

Here $n = 6$, which is even, so the rose has $2n$ petals: 12 petals.

- 24 How many petals does the rose $r = 4\cos(7\theta)$ have?

- a. 14
- b. 9
- c. 7 ✓
- d. 5

Here $n = 7$, which is odd, so the rose has exactly n petals: 7 petals.

25 How many petals does the rose $r = 4\cos(2\theta)$ have?

- a. 7
- b. 2
- c. 4 ✓
- d. 6

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

26 How many petals does the rose $r = 6\cos(2\theta)$ have?

- a. 7
- b. 4 ✓
- c. 2
- d. 6

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

27 How many petals does the rose $r = 3\sin(7\theta)$ have?

- a. 9
- b. 7 ✓
- c. 14
- d. 5

Here $n = 7$, which is odd, so the rose has exactly n petals: 7 petals.

28 How many petals does the rose $r = 6\sin(7\theta)$ have?

- a. 14
- b. 5
- c. 7 ✓
- d. 9

Here $n = 7$, which is odd, so the rose has exactly n petals: 7 petals.

29 How many petals does the rose $r = 9\cos(4\theta)$ have?

a. 8 ✓

b. 6

c. 4

d. 10

Here $n = 4$, which is even, so the rose has $2n$ petals: 8 petals.

30 How many petals does the rose $r = 3\sin(4\theta)$ have?

a. 6

b. 4

c. 10

d. 8 ✓

Here $n = 4$, which is even, so the rose has $2n$ petals: 8 petals.

31 How many petals does the rose $r = 2\sin(3\theta)$ have?

a. 6

b. 5

c. 1

d. 3 ✓

Here $n = 3$, which is odd, so the rose has exactly n petals: 3 petals.

32 How many petals does the rose $r = 5\cos(2\theta)$ have?

a. 6

b. 4 ✓

c. 1

d. 2

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

33 How many petals does the rose $r = 2\cos(4\theta)$ have?

a. 6

b. 4

c. 10

d. 8 ✓

Here $n = 4$, which is even, so the rose has $2n$ petals: 8 petals.

34 How many petals does the rose $r = 2\cos(8\theta)$ have?

a. 14

b. 18

c. 16 ✓

d. 8

Here $n = 8$, which is even, so the rose has $2n$ petals: 16 petals.

35 How many petals does the rose $r = 2\sin(6\theta)$ have?

a. 14

b. 12 ✓

c. 10

d. 6

Here $n = 6$, which is even, so the rose has $2n$ petals: 12 petals.

36 How many petals does the rose $r = 7\cos(7\theta)$ have?

- a. 5
- b. 9
- c. 7 ✓
- d. 14

Here $n = 7$, which is odd, so the rose has exactly n petals: 7 petals.

Multiple choice — medium 15

37 Convert the polar point $(-8, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(-1, 10)$
- b. $(8, 0)$
- c. $(0, 8)$ ✓
- d. $(0, -8)$

$$x = r\cos\theta = -8\cos\left(\frac{3\pi}{2}\right) = 0. \quad y = r\sin\theta = -8\sin\left(\frac{3\pi}{2}\right) = 8. \quad \text{Point: } (0, 8).$$

38 Convert the polar point $(-2, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(0, 2)$ ✓
- b. $(2, 0)$
- c. $(-2, 3)$
- d. $(0, -2)$

$$x = r\cos\theta = -2\cos\left(\frac{3\pi}{2}\right) = 0. \quad y = r\sin\theta = -2\sin\left(\frac{3\pi}{2}\right) = 2. \quad \text{Point: } (0, 2).$$

39 Convert the polar point $(6, \pi)$ to rectangular coordinates.

- a. $(0, -6)$
- b. $(6, 0)$
- c. $(-6, 0)$ ✓
- d. $(-5, 0)$

$$x = r \cos \theta = 6 \cos (\pi) = -6. \quad y = r \sin \theta = 6 \sin (\pi) = 0. \quad \text{Point: } (-6, 0).$$

40 Convert the polar point $(2, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(2, 0)$
- b. $(0, 2)$ ✓
- c. $(0, -2)$
- d. $(2, 4)$

$$x = r \cos \theta = 2 \cos \left(\frac{\pi}{2}\right) = 0. \quad y = r \sin \theta = 2 \sin \left(\frac{\pi}{2}\right) = 2. \quad \text{Point: } (0, 2).$$

41 Convert the polar point $(-8, \pi)$ to rectangular coordinates.

- a. $(9, 0)$
- b. $(0, 8)$
- c. $(-8, 0)$
- d. $(8, 0)$ ✓

$$x = r \cos \theta = -8 \cos (\pi) = 8. \quad y = r \sin \theta = -8 \sin (\pi) = 0. \quad \text{Point: } (8, 0).$$

42 Convert the polar point $(-5, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(5, 0)$
- b. $(0, 5)$ ✓
- c. $(0, -5)$
- d. $(2, 3)$

$$x = r \cos \theta = -5 \cos \left(\frac{3\pi}{2}\right) = 0. \quad y = r \sin \theta = -5 \sin \left(\frac{3\pi}{2}\right) = 5. \quad \text{Point: } (0, 5).$$

43 Convert the polar point $(7, \pi)$ to rectangular coordinates.

- a. $(7, 0)$
- b. $(0, -7)$
- c. $(-6, 0)$
- d. $(-7, 0)$ ✓

$$x = r \cos \theta = 7 \cos (\pi) = -7. \quad y = r \sin \theta = 7 \sin (\pi) = 0. \quad \text{Point: } (-7, 0).$$

44 Convert the polar point $(2, \pi)$ to rectangular coordinates.

- a. $(2, 0)$
- b. $(-2, 0)$ ✓
- c. $(-1, 0)$
- d. $(0, -2)$

$$x = r \cos \theta = 2 \cos (\pi) = -2. \quad y = r \sin \theta = 2 \sin (\pi) = 0. \quad \text{Point: } (-2, 0).$$

45 Convert the polar point $(4, 0)$ to rectangular coordinates.

- a. $(0, 4)$
- b. $(5, 0)$
- c. $(-4, 0)$
- d. $(4, 0)$ ✓

$$x = r \cos \theta = 4 \cos (0) = 4. \quad y = r \sin \theta = 4 \sin (0) = 0. \quad \text{Point: } (4, 0).$$

46 Convert the polar point $(5, 0)$ to rectangular coordinates.

- a. $(6, 0)$
- b. $(-5, 0)$
- c. $(5, 0)$ ✓
- d. $(0, 5)$

$$x = r \cos \theta = 5 \cos(0) = 5. \quad y = r \sin \theta = 5 \sin(0) = 0. \quad \text{Point: } (5, 0).$$

47 Convert the polar point $(-3, 0)$ to rectangular coordinates.

- a. $(-3, 0)$ ✓
- b. $(3, 0)$
- c. $(-2, 0)$
- d. $(0, -3)$

$$x = r \cos \theta = -3 \cos(0) = -3. \quad y = r \sin \theta = -3 \sin(0) = 0. \quad \text{Point: } (-3, 0).$$

48 Convert the polar point $(4, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(0, -4)$
- b. $(1, 3)$
- c. $(0, 4)$ ✓
- d. $(4, 0)$

$$x = r \cos \theta = 4 \cos(\frac{\pi}{2}) = 0. \quad y = r \sin \theta = 4 \sin(\frac{\pi}{2}) = 4. \quad \text{Point: } (0, 4).$$

49 Convert the polar point $(9, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(0, 9)$ ✓
- b. $(0, -9)$
- c. $(9, 0)$
- d. $(2, 11)$

$$x = r \cos \theta = 9 \cos(\frac{\pi}{2}) = 0. \quad y = r \sin \theta = 9 \sin(\frac{\pi}{2}) = 9. \quad \text{Point: } (0, 9).$$

50 Convert the polar point $(-6, \pi)$ to rectangular coordinates.

- a. $(0, 6)$
- b. $(-6, 0)$
- c. $(7, 0)$
- d. $(6, 0)$ ✓

$$x = r \cos \theta = -6 \cos(\pi) = 6. \quad y = r \sin \theta = -6 \sin(\pi) = 0. \quad \text{Point: } (6, 0).$$

51 Convert the polar point $(-5, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(-5, 0)$
- b. $(0, -5)$ ✓
- c. $(0, 5)$
- d. $(2, -3)$

$$x = r \cos \theta = -5 \cos\left(\frac{\pi}{2}\right) = 0. \quad y = r \sin \theta = -5 \sin\left(\frac{\pi}{2}\right) = -5. \quad \text{Point: } (0, -5).$$

Multiple choice — hard 15

52 Classify the curve $r = \sin \theta$.

- a. Cardioid
- b. Limaçon without an inner loop
- c. Circle through the pole ✓
- d. Limaçon with an inner loop

For $r = \sin \theta$: the constant term is 0, so it passes through the pole and traces a circle. This is a circle through the pole.

53 Classify the curve $r = 9 + 9\cos \theta$.

- a. Circle through the pole
- b. Limaçon with an inner loop
- c. Limaçon without an inner loop
- d. **Cardioid ✓**

For $r = 9 + 9\cos \theta$: the constant term equals the coefficient, the defining property of a cardioid. This is a cardioid.

54 Classify the curve $r = 3 + 5\sin \theta$.

- a. Limaçon without an inner loop
- b. Circle through the pole
- c. Cardioid
- d. **Limaçon with an inner loop ✓**

For $r = 3 + 5\sin \theta$: the constant term is less than the coefficient, so $r = 0$ for some θ and the curve loops back through the pole. This is a limaçon with an inner loop.

55 Classify the curve $r = 7 + 5\cos \theta$.

- a. Circle through the pole
- b. **Limaçon without an inner loop ✓**
- c. Limaçon with an inner loop
- d. Cardioid

For $r = 7 + 5\cos \theta$: the constant term exceeds the coefficient, so r never reaches 0 and there is no inner loop. This is a limaçon without an inner loop.

56 Classify the curve $r = 5 + 5\sin \theta$.

- a. Limaçon without an inner loop
- b. Cardioid ✓**
- c. Circle through the pole
- d. Limaçon with an inner loop

For $r = 5 + 5\sin \theta$: the constant term equals the coefficient, the defining property of a cardioid. This is a cardioid.

57 Find the smallest positive θ where the rose $r = 3\sin(4\theta)$ passes through the pole.

- a. $\frac{\pi}{4}$ ✓**
- b. $\frac{\pi}{16}$
- c. $\frac{3\pi}{4}$
- d. $\frac{\pi}{8}$

Set $3\sin(4\theta) = 0 \Rightarrow \sin(4\theta) = 0$. The smallest positive solution is $4\theta = \pi$, so $\theta = \frac{\pi}{4}$.

58 Find the smallest positive θ where the rose $r = 4\cos(6\theta)$ passes through the pole.

- a. $\frac{7\pi}{12}$
- b. $\frac{\pi}{24}$
- c. $\frac{\pi}{6}$
- d. $\frac{\pi}{12}$ ✓**

Set $4\cos(6\theta) = 0 \Rightarrow \cos(6\theta) = 0$. The smallest positive solution is $6\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{12}$.

59 Find the smallest positive θ where the rose $r = 4\sin(4\theta)$ passes through the pole.

- a. $\frac{\pi}{16}$
- b. $\frac{\pi}{4}$ ✓
- c. $\frac{\pi}{8}$
- d. $\frac{3\pi}{4}$

Set $4\sin(4\theta) = 0 \Rightarrow \sin(4\theta) = 0$. The smallest positive solution is $4\theta = \pi$, so $\theta = \frac{\pi}{4}$.

60 Find the smallest positive θ where the rose $r = 9\cos(4\theta)$ passes through the pole.

- a. $\frac{\pi}{16}$
- b. $\frac{\pi}{4}$
- c. $\frac{5\pi}{8}$
- d. $\frac{\pi}{8}$ ✓

Set $9\cos(4\theta) = 0 \Rightarrow \cos(4\theta) = 0$. The smallest positive solution is $4\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{8}$.

61 Find the smallest positive θ where the rose $r = 8\cos(5\theta)$ passes through the pole.

- a. $\frac{3\pi}{5}$
- b. $\frac{\pi}{10}$ ✓
- c. $\frac{\pi}{5}$
- d. $\frac{\pi}{20}$

Set $8\cos(5\theta) = 0 \Rightarrow \cos(5\theta) = 0$. The smallest positive solution is $5\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{10}$.

62 The limaçon $r = 7 + 14\cos\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

- a. $\frac{2\pi}{3}$ ✓
- b. $\frac{\pi}{2}$
- c. $\frac{5\pi}{6}$
- d. $\frac{4\pi}{3}$

Set $r = 0$: $7 + 14\cos\theta = 0 \Rightarrow \cos\theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{2\pi}{3}$.

63 The limaçon $r = 7 + 14\sin \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{7\pi}{6}$ ✓

b. $\frac{11\pi}{6}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{2}$

Set $r = 0$: $7 + 14\sin \theta = 0 \Rightarrow \sin \theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{7\pi}{6}$.

64 The limaçon $r = 3 + 6\cos \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{2\pi}{3}$ ✓

b. $\frac{5\pi}{6}$

c. $\frac{\pi}{2}$

d. $\frac{4\pi}{3}$

Set $r = 0$: $3 + 6\cos \theta = 0 \Rightarrow \cos \theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{2\pi}{3}$.

65 The limaçon $r = 8 + 16\sin \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{5\pi}{6}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$ ✓

d. $\frac{11\pi}{6}$

Set $r = 0$: $8 + 16\sin \theta = 0 \Rightarrow \sin \theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{7\pi}{6}$.

66 The limaçon $r = 1 + 2\cos \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{2\pi}{3}$ ✓

b. $\frac{\pi}{2}$

c. $\frac{4\pi}{3}$

d. $\frac{5\pi}{6}$

Set $r = 0$: $1 + 2\cos \theta = 0 \Rightarrow \cos \theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{2\pi}{3}$.